Fakhreddin (Dean) Emami

Columbia, South Carolina (Willing to relocate) fakhreddinemami@gmail.com 517 940 5979
Dean-emami Google Scholar

Summary

FEA Simulation Engineer | FEA & AI Integration | HPC/ML Surrogates

FEA Simulation Engineer with extensive experience in computational solid mechanics, high-fidelity finite element analysis (FEA), optimization, and data-driven modeling. Skilled in developing large-scale computational and surrogate-modeling platforms using machine learning and high- performance computing (HPC) to advance the design, analysis, and characterization of complex materials and structural systems. Proven track record of contributing to NASA and Department of Defense (DARPA/DoD) projects, with a strong focus on mechanics of materials, inverse identification of constitutive models, stress and structural analysis, and scalable computational frameworks for complex engineering systems.

Core Expertise: FEA · Nonlinear Material Modeling · Structural/Stress Analysis · Surrogate Modeling · Optimization · SciML · HPC/GPU · Python · Abaqus

Professional Experience

FEA Simulation Researcher — Aug. 2021 – Present
University of South Carolina — Columbia, SC

- Led **FEA, multi-physics simulation**, ML, and experimental workflows for $500k **DARPA** and $100k **NASA** programs, coordinating graduate and undergraduate researchers and delivering validated simulation models and materials-calibration milestones.
- Executed **1M+** high-fidelity **ABAQUS**/UMAT/USDFLD **simulations** using Python/Fortran automation, performing **static**, **dynamic**, **buckling**, **modal**, and **thermal analyses** as well as submodeling and mesh-convergence verification, generating datasets that enabled NASA's Venus-buoyant architected-materials design and DARPA's single-test inverse calibration.
- Developed large-scale scientific **ML surrogates**—including shared-weight **MLPs** and **GNNs**—trained on **27 TB** of synthetic FEA **data**, producing high-fidelity inverse-calibration error functions with integrated uncertainty quantification (**UQ**) and reducing full FEA workflows from **300 CPU-hours to 5 minutes** using GPU accelerated inference.
- Engineered **CPU/GPU**-accelerated **HPC pipelines** with **Python**, **CUDA/CuPy**, **PyTorch**, and ABAQUS, implementing **automated** submission, analysis, and post-processing with license-aware scheduling and adaptive submission-rate control, supporting 1M+ reproducible simulations across HPC clusters.
- Developed a DARPA-funded 3D heterogeneous specimen enabling single-test **inverse calibration of metal plasticity** parameters, replacing 12 conventional tests, eliminating bias and non-uniqueness, and reducing calibration time from weeks to hours.
- Performed **material and geometry optimization** for NASA's architected-materials program, generating ceramic lattice designs capable of surviving Venus-level **mechanical and thermal loads** while providing 43 kg/m^3 buoyancy and thermal conductivity as low as 0.14 W/m·K.
- Designed optimized 2D and 3D architected-material and structural systems using analytical and numerical modeling, achieving **2×–6× improvements in stiffness and strength**.
- Provided cross-functional technical leadership, mentoring graduate and undergraduate researchers and coordinating simulation and DIC experimental workflows; contributed to **10+ publications**, national conference presentations, and a **U.S. patent**.

Structural/Civil Engineer — Mar. 2020 – Jul. 2021
Saeed Associates Chartered — Springfield, VA

- Performed structural analysis, calculations, and **design** for **highway and bridge** components in compliance with AASHTO, VDOT, AISC, ACI, PCA, and other relevant codes, delivering dependable designs for major infrastructure projects including HRBT and I-66.
- Developed structural and civil **CAD** drawing using MicroStation for I-66 and HRBT project in Virginia, resulting in professional sheets before deadlines.
- Completed **quantity estimates** and reviewed contractor **shop drawings**, identifying discrepancies and ensuring code-compliant construction documentation, which reduced field changes and rework.

Education

Ph.D. Mechanical Engineering, University of South Carolina, Columbia, SC, USA — Aug. 2021 – May 2026
M.Sc. Civil Engineering , Michigan State University, East Lansing, MI, USA — Aug. 2018 – Dec. 2019
B.Sc. Civil Engineering , Azad University Central Tehran Branch, Tehran, Iran — Aug. 2010 – May 2014

Technical Skills

Technical Competencies: Finite Element Analysis (FEA) · Thermo-Mechanical Simulation · Nonlinear Modeling · Inverse Calibration · Surrogate Modeling · Optimization (CMA-ES, gradient-free) · Uncertainty Quantification (UQ) · Computational Solid Mechanics · Structural Analysis (static, dynamic, buckling, modal, thermal) · Data-Driven Modeling · High-Performance Computing (HPC) · Machine Learning–Driven Simulation · Material Behavior Characterization

Programming and HPC: Python · Fortran · MATLAB · CUDA/CuPy · PyTorch · HPC Workflows · Parallel Computing · GPU-Accelerated Computing · Scripting & Automation

Engineering Software: ABAQUS (UMAT/USDFLD) · LS-DYNA · ETABS · SAFE · SolidWorks · Fusion 360 · MicroStation · AutoCAD · Meshmixer

Patent

- **Emami** & Gross — Ceramic kelvin foam with geometry optimized for hydrostatic loading", US Patent 12,415,594.

Certificates

- Fundamentals of Engineering Examination (EIT)

Publications

- **Emami** & Gross — Making Latent Physics Explicit: An AI-Enabled Surrogate Error Function (Ongoing - 2026)
- **Emami** et al.— AI-Enhanced Inverse Calibration of Plasticity Models Using a 3D Heterogeneous Specimens (Ongoing - 2026)
- **Emami** et al.— A New Performance Indicator for Heterogeneous Elastoplastic Material Test Specimens (IJMS - 2025)
- **Emami** & Gross — Mechanical Performance of Triangular Lattices: The Role of Node Fillets (IJSS - 2025)
- **Emami** et al.— Estimation of Hierarchical Honeycomb Mechanical Properties By a Knowledge-Extraction-Based Surrogate Model (AIAA 2025)
- **Emami** & Gross —Better than linear strength scaling of multifunctional ceramic truss lattice materials (IJMS - 2024)
- **Emami** & Gross — Warren truss inspired hierarchical beams for three dimensional hierarchical truss lattice materials (MoM 2024)
- **Emami** & Gross — Design of Buoyant Architected Materials to Enable a New Aerial Platform Operating Near the Surface of Venus (AIAA 2023)
- **Emami** & Gross — Mechanical Properties of Hierarchical Beams for Large-Scale Space Structures (AIAA 2023)
- **Emami** & Kabir — Strength prediction of composite metal deck slabs under free drop impact loading using FEM and machine learning approach (SI 2023)
- **Emami** & Kabir — Performance of composite metal deck slabs under impact loading (JoS 2019)